643557

Course Name: Introduction to Microprocessor

Course Outcome

CO1- Demonstrate the Microprocessor internal architecture and its operations.
CO2- Develop programs based on 8085 microprocessor instruction set and addressing mode.
CO3- Develop program using looping, counting, indexing, counter and time delays.
CO4- Understand the concept of stack and subroutine for modular approach.
CO5- Compare accepted standards and guidelines to select microprocessor (8085 and 8086) to meet performance requirements.
CO6- Analyze the concept of interfacing the processor to external device with I/O programming and Interrupt Driven I/O.
CO7- Understand the working of interfacing chips (8257/37, 8253/54, 8255 & 8259).

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End Term Examination, Even Semester 2023-24
B. Tech (CSE/AIML/DA/IIOT/CCV/CSF), II Year, IV Semester
INTRODUCTION TO MICROPROCESSOR (BCSC 0007)

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. All parts of a question should be answered at one place.
2. First complete the one set of sections then start a new section.
3. Assume the appropriate data for any missing data.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Observe and write the contents of the Accumulator and the Processor Status Word (PSW) (in Hexadecimal format) after the execution of the following sequence of instructions. Assume [2000H] = 0AH. STC MVI B, 5CH LXI D, 2000H LDAX D RAR ADD B HLT	4	2	U	C
2	Distinguish the following instructions: (a) SIM and RIM (b) RAL and RLC (c) LDAX and STAX (d) IN and OUT	4	2	R	F

3	Discuss the concept of Stack in 8085. Also, exchange the contents of BC and DE register pairs using stack instructions only.	4	4	U	C
4	A team of embedded design engineers has selected 8086 to speed up the execution of the instructions. How the team would have observed the superiority of 8086 over the 8085 microprocessor on the grounds of architecture, key components, and pipelining features?	4	1	U	F
5	A basic communication system has to be set up between the machine and the microprocessor through 8255 PPI. Being a system engineer draw the internal architecture of 8255. Also, list the various operating modes of 8255. Specify the bit of a control word for 8255 which differentiates between I/O mode and BSR mode. OR A project manager has to lead the development of a cutting-edge autonomous vehicle control system. The assigned team has to simulate some critical tasks like real-time sensor data processing and emergency response coordination with the help of the Programmable Interrupt Controller (Intel 8259 PIC). How does the PIC enhance the system's ability by handling these demanding tasks efficiently and reliably?	4	7	U	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Compare and contrast the I/O mapped I/O and memory mapped I/O.	3	6	R	F
7	Write instructions to add the content of the memory location 2040H to Accumulator and subtract the content of the memory location 2041H from the first sum. Also, specify the final content of the Accumulator. Assume [A] = 30H, [2040H] = 68H and [2041H] = 7FH	3	2	U	P

8	Write an assembly language program for 8085 to find the smallest element in a block of 8-bit unsigned binary data. The block begins from A002H and the length of the block is stored at the memory location A001H. Also, store the smallest number at the memory location FF00H.	3	2	A	P
9	For an 8085-8255-based microcomputer, write a *program to read the DIP switches and display the* reading from port B at port A and from port C_L at port C_U. Let the address of the Control Word Register (CWR) be 8003H.	3	6	U	P
10	A graduate engineer trainee wrote the following assembly language program on an 8085 microprocessor to compute the precise delay while working on a real-time control system project. How the trainee will implement a loop to achieve the delay? MVI B, 13H LOOP2: MVI C, FDH LOOP1: DCR C JNZ LOOP1 DCR B JNZ LOOP2 Assume the microprocessor is operating at the frequency of 2 MHz.	3	3	U	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Write a program to count from '0' to '9' with 1 second of delay between each count. At the count of 9, the counter should reset itself to '0' and repeat the sequence continuously. Use the register pair DE to set up the delay and display each count at one of the output ports. Also, draw a flow chart to show the	5	3	A	P

	calculation of setting 1 second delay. Assume the clock frequency of the microcomputer is 1 MHz.				
12	Draw the conditional and control bit pattern for the flag register of 8086 and discuss the significance of each bit. Explain the concept of memory segmentation in the 8086 microprocessor with a suitable illustration. Also, list the different addressing modes of 8086 with suitable examples.	5	5	U	F
13	Consider, that a system has been designed for a high-speed data acquisition application where data from multiple sensors needs to be continuously transferred to memory for processing. Describe how the 8257 DMA controller will be utilized to efficiently handle these data transfers.	5	7	U	F